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Time-variable gravity potential components for optical clock comparisons and the definition of international time scales

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Abstract

The latest generation of optical atomic clocks is approaching the level of one part in 10^{18} in terms of frequency stability and uncertainty. For clock comparisons and the definition of international time scales, a relativistic redshift effect of the clock frequencies has to be taken into account at a corresponding uncertainty level of about 0.1 $m^2\ s^{-2}$ and 0.01 m in terms of gravity potential and height, respectively. Besides the predominant static part of the gravity potential, temporal variations must be considered in order to avoid systematic frequency shifts. Time-variable gravity potential components induced by tides and non-tidal mass redistributions are investigated with regard to the level of one part in 10^{18}. The magnitudes and dominant time periods of the individual gravity potential contributions are investigated globally and for specific laboratory sites together with the related uncertainty estimates. The basics of the computation methods are presented along with the applied models, data sets and software. Solid Earth tides contribute by far the most dominant signal with a global maximum amplitude of 4.2 $m^2\ s^{-2}$ for the potential and a range (maximum-to-minimum) of up to 1.3 and 10.0 $m^2\ s^{-2}$ in terms of potential differences between specific laboratories over continental and intercontinental scales, respectively. Amplitudes of the ocean tidal loading potential can amount up to 1.25 $m^2\ s^{-2}$, while the range of the potential between specific laboratories is 0.3 and 1.1 $m^2\ s^{-2}$ over continental and intercontinental scales, respectively. These are the only two contributors being relevant at a 10^{-17} level. However, several other time-variable potential effects can particularly affect clock comparisons at the 10^{-18} level. Besides solid Earth pole tides, these are non-tidal mass redistributions in the atmosphere, the oceans and the continental water storage.

Keywords: gravity potential, relativistic redshift effect, definition of time scales, optical clocks, clock comparisons

(Some figures may appear in colour only in the online journal)

1. Introduction

Over the past decade, various laboratories around the world have developed prototype optical atomic clocks, approaching fractional stabilities and uncertainties of one part in 10^{18} [1–6]. The utilization of such clocks for contributions to international time scales and applications in fundamental physics, geodesy, navigation, etc, require local and remote comparisons over long distances [7, 8]. At present, only optical fibre links have demonstrated frequency transfer performances below one part in 10^{18} over short, medium and continental distances up to a few 1000 km [9–16], while GNSS (Global Navigation Satellite Systems) and other satellite frequency transfer techniques, including two-way procedures such as TWSTFT (two-way

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satellite time and frequency transfer), perform roughly two orders of magnitude below optical fibres [17].

In general, the definition of time scales and clock comparison procedures must be handled within the framework of general relativity, where the effects must be evaluated at an improved level of accuracy for the optical clocks. These effects include the relativistic redshift, the velocity of the clocks, and the Sagnac term. At the level of one part in 10^{18}, the evaluation of the relativistic effects for an earthbound clock at rest require only the gravity potential (gravitational plus centrifugal component), while for a transported clock also the velocity with respect to the Earth must be considered, leading to the Sagnac term. In this context, systematic relativistic treatments of time and frequency transfer in the vicinity of the Earth (including satellites) can be found in [18–22], while the situation for optical fibres is discussed in detail in [23].

According to the laws of general relativity, one part in 10^{18} clock performance corresponds to about 0.1 $m^2\,s^{-2}$ in terms of the gravity potential, which is equivalent to 0.01 m in height. Therefore, it is necessary to determine the gravity potential with corresponding accuracy at each clock site for the time of the experiment, as the gravity potential also includes time-variable components. Hence, besides the predominant static part of the gravity potential, also the time-variable components must be evaluated, which are dominated by (mainly periodic) tidal effects, reaching amplitudes of about 4 $m^2\,s^{-2}$. Regarding future contributions of optical clocks to international time scales, the absolute gravity potential values must be known relative to a given conventional reference potential, while the knowledge of corresponding gravity potential differences is sufficient for pure clock comparisons. With respect to the magnitude of the corrections for time-variable components of the gravity potential this means that the amplitudes are relevant for contributions to international time scales, while for clock comparisons between two stations the corresponding difference signal matters, which will be generally small and almost cancel out over short distances, but may reach twice the amplitude of the corresponding station time series over longer (intercontinental) distances due to the usually occurring phase shifts.

In the context of clock comparisons and time scales, the issue of time variations in the gravity potential is known in principle and was addressed already in [7, 24], and more detailed discussions of individual effects can be found, e.g. in [21]. However, a comprehensive study of the magnitudes and dominant time periods of time-variable gravity potential effects with regard to clock comparisons and time scales at a level of one part in 10^{18} is lacking so far. Therefore, the main objective of this contribution is the analysis of the relevant effects, including the dominant solid Earth and ocean tide contributions as well as other tidal effects, e.g. from polar motion, but also non-tidal mass redistributions in the atmosphere, the oceans and the continental water storage. The global maximum amplitudes and dominant time periods of the gravity potential effects are computed. The corresponding difference signals between specific laboratory sites are investigated, aiming at contributions to international time scales and clock comparisons, respectively. The associated uncertainties are estimated as well. The basics of the computation methods are presented along with the applied models, data sets and state-of-the-art software.

The entire work was inspired by the ITOC (international time scales with optical clocks) project [7] funded by the EMRP (European metrology research programme). The ITOC project is a collaborative European effort to address the key challenges for a possible redefinition of the SI second, and includes a coordinated programme for the comparison of optical clocks developed at five different national metrological institutes (NMIs), a proof-of-principle experiment to demonstrate the determination of the gravity potential by optical clocks, as well as the development of the necessary relativistic formulations. Within the ITOC project, a so-called researcher excellence grant (REG) was given to the Leibniz Universität Hannover (LUH), being responsible for all tasks related to the gravity potential determination for the optical clock experiments.

This article begins with the relativistic background and defines the general terms for optical clock comparisons and the definition of international time scales (chapter 2). Chapter 3 provides the time-variable components of the gravity potential. It starts with a section on general aspects, which apply for all effects, followed by individual sections for the time-variable components that are known to be relevant in high precision geodesy. Finally, the results are summarized, and their significance is discussed.

2. Relativistic background

Einstein's general relativity theory (GRT) predicts that two ideal clocks will in general run at different rates if they move with a relative velocity or are under the influence of different gravitational fields, which is associated with the relativistic redshift effect (one of the classical general relativity tests). In order to describe time and frequency comparisons and to construct and disseminate international time scales, a proper relativistic reference system has to be chosen first. In this context, the (spacetime) geocentric celestial reference system (GCRS) with the geocentric coordinate time (TCG) is most suitable for all experiments and observations in the vicinity of the Earth. The GCRS is a 'space-fixed' geocentric system, which is kinematically non-rotating with respect to extragalactic radio sources (quasars); the system and the associated metric is defined and recommended within resolutions passed by the International Astronomical Union (IAU) in the year 2000. In this framework, the relation between proper (observed) time τ and coordinate time t is given by

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = 1 - \frac{V}{c^2} - \frac{v^2}{2c^2} + o(c^{-4}), \qquad (1)$$

where V is the (Newtonian) gravitational scalar potential, including time-variable components such as tidal effects, v is the coordinate velocity of the observer in the (non-rotating) GCRS, and c is the speed of light in vacuum. The above equation is accurate to about 5 parts in 10^{19} for locations from the Earth's surface up to geostationary orbits, provided that

time-variable components in V are taken into account; this is at present fully sufficient in the context of optical clocks (for further details see [25]). The potential V is defined with positive sign, following generally accepted geodetic practice. This leads to the fact that the proper time interval $d\tau$, separating two events, is less than the corresponding coordinate time interval dt. Consequently, clocks run slower (tick slower, show less time) when they move or are affected by gravitation; this slowing of time is called time dilation, and may be separated into a (relative) velocity and a gravitational time dilation, or sometimes also called special and general relativistic time dilation, respectively.

When switching from the GCRS to an Earth-fixed reference system, co-rotating with the Earth, and assuming an observer (clock) at rest within this system, the velocity component in (1) is purely related to Earth rotation, given by $v = \omega p$ with the angular velocity ω about the Earth's rotation axis and the distance p from the rotation axis. Hence, the term $v^2/2$ in (1) is exactly the centrifugal potential $Z = \omega^2 p^2/2$ and the relation between proper and coordinate time can be written as

$$\frac{d\tau}{dt} = 1 - \frac{W}{c^2} + o(c^{-4}), \qquad (2)$$

where $W = V + Z$ is the gravity (gravitational plus centrifugal) potential, commonly used in geodesy. If the observer is not at rest, two more terms enter in the above equation, one related to the relative velocity in the Earth-fixed system and the second one being the Sagnac term [26].

The above relations between proper and coordinate time form the basis for the relativistic treatment of time and frequency transfers, as outlined in detail in [18–22], and they also provide the link to the relativistic redshift effect, which considers an electromagnetic wave (light) travelling from an emitter (located at P_0) to a receiver (located at P) in conjunction with two ideal (error free) frequency standards (clocks) located at P and P_0, respectively. This leads to the relation

$$\frac{\Delta f}{f_P} = \frac{f_P - f_0}{f_P} = 1 - \frac{d\tau_P}{d\tau_0} = \frac{W_P - W_0}{c^2} + o(c^{-4}) \approx -gH, \qquad (3)$$

where f_P and f_0 are the proper frequencies of the transmitted signal as measured at points P and P_0, respectively, while W_P and W_0 are the corresponding gravity potential values, and H is the vertical distance of point P relative to point P_0, measured positive upward. Assuming that P is located above P_0, $W_P - W_0$ is negative and so is Δf, i.e. the received (observed) frequency f_P is lower than the emitted frequency f_0, or in other words, the observer at P will find that the (lower) clock at P_0 is running slow with respect to his own clock, and thus blue light becomes more red, explaining the denotation 'redshift effect'. In this context, the term 'relativistic redshift effect' is preferred over 'gravitational redshift effect', as for earth-bound clocks not just the gravitational but also the centrifugal potential is involved, with similar conclusions being drawn by [20, 27, 28].

For the construction and dissemination of international time scales, terrestrial time (TT) was introduced for all practical timing issues at or near the Earth's surface. TT is another coordinate time attached to the GCRS, which differs from TCG by just a constant rate [29]

$$\frac{dTT}{dTCG} = 1 - L_G, \qquad L_G = 6.969\,290\,134 \times 10^{-10}. \qquad (4)$$

The main idea behind TT is to produce a time scale, which is associated with an ideal hypothetical atomic clock located on the rotating geoid as a selected (zero) level surface within the Earth's gravity field. In the original definition of TT it was mentioned that the unit of measurement should agree with the 'SI second on the rotating geoid', but due to the intricacy and problems associated with the definition, realization and changes of the geoid, the IAU decided in the year 2000 to turn L_G into a defining (error free) constant with a fixed value, thus avoiding the explicit mention of the geoid. The numerical value of L_G was chosen to maintain continuity with the previous definition, and as the relation $L_G = W_0/c^2$ holds according to (2), this implicitly defines a zero gravity potential value with $W_0 = 62\,636\,856.00$ m^2 s^{-2}. This zero gravity potential value W_0 is also listed in the IERS conventions 2010 [29], and consequently, in order to be consistent with the IERS and the even more important IAU recommendations, it is reasonable to choose this value as the reference potential for all applications related to international time scales. A further complication in this context arises from the fact that the International Association of Geodesy (IAG) officially adopted in its Resolution No. 1 from 2015 a value of $W_0 = 62\,636\,853.4$ m^2 s^{-2} according to [30], and the consequences of the two conflicting values W_0 should be resolved in the future by international cooperation.

TT is also a theoretical coordinate time scale, and thus it can have different realizations, such as TAI (international atomic time), UTC (coordinated universal time), GNSS time systems, etc, differing by specific offsets. Therefore, in order to contribute to these international time scales, the individual clocks require a correction for relativistic effects, aiming at the generation of time and frequency signals associated with a hypothetical clock located on the (rotating) geoid, or more precisely the zero level surface associated with the gravity potential $W = W_0$. As already discussed above, a clock located at P above the zero level surface W_0 will run faster (tick faster, show more time) than a corresponding clock at P_0 somewhere on the zero level surface, which is directly connected to the relativistic redshift effect. Consequently, the clock frequency or clock rate at P above the zero level surface must be reduced (lowered) by an amount Δf, given by (3), in order to produce a signal corresponding to a hypothetical clock at the zero level surface, or in other words, the clock frequency at P has to be adjusted such that it will no longer observe a redshift of the clock signal emitted at P_0. As all existing primary frequency standards are situated above the geoid, the relativistic redshift corrections can become quite significant, e.g. [28] estimated a correction $\Delta f/f$ of about -1800×10^{-16} for NIST (National Institute of Standards and Technology, United States) at Boulder, Colorado, with an altitude of about 1650 m.

In summary, relativistic redshift corrections according to (3) must be computed for all clocks contributing to international time scales, the most important ones being TAI and

UTC as realizations of TT. In this context, absolute gravity potential values W are required first, which then have to be referenced to a conventional zero potential value W_0. On the other hand, pure local and remote clock comparisons only require corresponding gravity potential differences between the relevant clock sites without the need to know the absolute potential values. With regard to the orders of magnitude, (3) directly shows that one part in 10^{18} clock performance corresponds to about 0.1 $m^2\ s^{-2}$ in terms of the gravity potential, which is equivalent to 0.01 m in height. Consequently, the gravity potential (or differences thereof) should be determined with corresponding accuracy for all clock sites, or vice versa, optical clocks may be used in the future to directly measure gravity potential differences, this technique being an important application within the framework of a relativistic geodesy, also denoted as chronometric levelling or chronometric geodesy (e.g. [27, 31–33]).

For clock performances at the level of one part in 10^{18}, equivalent to 0.1 $m^2\ s^{-2}$ in terms of the gravity potential, also time-variable components of the potential function W must be considered for the times when clock observations take place. Hence, besides the predominant static part of the gravity potential, also the time-variable components need to be addressed, leading to

$$W(t) = W^{\text{static}} + W^{\text{temp}}. \tag{5}$$

The static part of the gravity potential can be derived in principle from geometric levelling on the one hand and GNSS positions in combination with a gravimetric geoid model on the other hand. Regarding possible accuracies, the most direct and accurate way to derive potential differences over short distances is the geometric levelling technique, as it can deliver potential differences at the level of 0.01 $m^2\ s^{-2}$ (0.001 m in height) and better. However, geometric levelling is susceptible to systematic errors, which may accumulate to the level of several 0.1 m over a few 1000 km. Therefore, over long distances the GNSS/geoid approach should be beneficial, allowing also the derivation of absolute potential values with uncertainties of about 0.2–0.3 $m^2\ s^{-2}$, and giving corresponding potential differences with uncertainties of about 0.1–0.3 $m^2\ s^{-2}$ on national scales, e.g. in Germany [34], and 0.2–0.5 $m^2\ s^{-2}$ on continental scales, e.g. in Europe [35], provided that high quality and resolution data are available within the area of interest. However, the emphasis in this contribution is on the temporal variations of the gravity field, aiming at a study of the magnitudes and dominant time periods of all relevant effects with regard to optical clocks operating at a level of one part in 10^{18}.

For contributions to international time scales, requiring the (absolute) gravity potential values relative to some given conventional reference potential, the amplitudes of the time-variable components play the main role, while for clock comparisons the corresponding difference signals matter, which will be generally small and almost cancel out over short distances, but may reach twice the amplitude of the corresponding station time series over longer (intercontinental) distances. Furthermore, as clock comparisons usually involve some averaging over time, also the maximum variation (maximum-to-minimum) of the difference signal (over time) is of interest. In this context, long distance optical clock comparisons, generally employing relatively short averaging times of one hour or less, require special attention, as in this case the time-variable gravity potential components may not average out sufficiently, while on the other hand, the tidal peak-to-peak signal may also serve for evaluating the performance of optical clocks (i.e. a detectability test). However, it should also be noted that intercontinental clock comparisons with short averaging times are presently limited in performance to the level of 10^{-16} (TWSTFT) to 10^{-17} (atomic clock ensemble in space ACES) until new technology emerges.

3. Time-variable components of the gravity potential field

3.1. General aspects

In the following sections, the effects of tidal signals and non-tidal mass redistributions on the gravity potential are investigated for stations on the deformable Earth's surface. Tidal forces arise from the gravitational acceleration of bodies external to the Earth, i.e. the Moon, the Sun and the planets of our solar system; they are usually taken as completely known and can be computed from the ephemerides of the external bodies and their masses, leading to the theoretical tides, equilibrium tidal potential or tide-generating potential (synonymous terms).

Based on an Earth model with knowledge about the elastic properties of the solid Earth, the convention of assuming an oceanless Earth (the effect of the ocean tides is put in at a later step), and some other restrictions, the Earth's response to the tidal potential can simply be described by the so-called Love numbers h_n, k_n, l_n [36], depending primarily on the spherical harmonic degree n of the tidal potential. For a tidal potential V_n at degree n and some surface location, a vertical (i.e. radial, positive upward) displacement of $h_n \times V_n/g$ (g is the Earth's gravity acceleration) and a horizontal displacement associated with the Love number l_n occur, while the distortion of the Earth produces an additional gravitational potential $k_n \times V_n$. The solid Earth model gives what are called the body tides, which are what would be observed on an oceanless but otherwise realistic Earth.

The mass fluctuations associated with the ocean tides cause a direct attraction effect of the corresponding water masses, which would lead to a change in the potential even on a rigid Earth, and a distortion of the Earth due to the loading effect. All these induced signals are called the load tides inducing the tidal loading potential V'_n. Similar to the external tidal forcing, the tidal loading effects as well as the effects of other surface mass loads can be computed by corresponding load Love numbers h'_n, k'_n, l'_n [37], and again the vertical and horizontal deformations are obtained by utilizing the factors h'_n and l'_n, respectively, while the additional potential due to the loading is obtained by employing the factor k'_n.

Now the direct and the indirect (deformation) effects lead to the total tidal potential $(1 + k_n) \times V_n$ and the total tidal

loading potential $(1 + k'_n) \times V'_n$ for the body and the load tides, respectively; this is the effect for a fixed point in space (referring to the centre of mass of the system Earth), as needed, e.g. for satellite geodesy applications. However, for points on the (solid, but deformable) Earth's surface (on which clocks, gravimeters, tide gauges, etc are located), it has to be considered that these points go up and down by $h_n \times V_n/g$ and $h'_n \times V'_n/g$, respectively, which results in an effective tidal potential $(1 + k_n - h_n) \times V_n$ and an effective tidal loading potential $(1 + k'_n - h'_n) \times V'_n$ for the body and load tides. Sometimes, the factor $\gamma = 1 + k_n - h_n$ is also called the diminishing factor, and accordingly this can be introduced for the load tides [38].

The temporal variations W^{temp} of the gravity potential in (5) are divided into tidal and non-tidal variations. Besides the Earth's body and ocean load tides, tidal variations (sections 3.2–3.6) also result from polar motion (solid Earth and ocean pole tides) and variations in the angular velocity of the Earth (length of day (LOD) tides) as well as from the atmosphere (atmospheric tides). The periodic part of the tidal variations can be modelled to a high degree of accuracy and is included in the IERS conventions [29] for the reduction of geodetic observations, but different options and definitions exist for the treatment of the permanent part of the solid Earth tides (section 3.2). Furthermore, the gravity potential is also affected by non-tidal mass redistributions in the atmosphere, oceans, hydrosphere, cryosphere and the solid Earth (crust, mantle and core), together with the induced deformations of the Earth (sections 3.7–3.11). These processes take place at different time scales and are of global, regional and local character. The non-tidal variations are so far not included in the IERS conventions 2010 due to the limited accuracy or lack of models at that time. However, such models are steadily improving with the satellite mission GRACE, being one of the main drivers on global scales. It should also be noted that the separation of individual contributions is in practice a difficult task, as the effects are more or less correlated in the real world.

The subsequent sections start with a general description of the individual phenomena, followed by an analysis of the amplitudes and dominant time periods of the gravity potential effects at specific laboratory sites for the reduction to a conventional zero potential value in the context of international time scales. Corresponding difference signals between laboratory sites are investigated in view of optical clock comparisons. Moreover, global analyses of corresponding potential effects are carried out. Regarding the relevant tidal effects, the national metrology institutes NPL (National Physical Laboratory, United Kingdom), PTB (Physikalisch-Technische Bundesanstalt, Germany) and NICT (National Institute of Information and Communication Technology, Japan) are chosen as typical example sites, and corresponding station time series as well as interstation difference time series over continental and intercontinental scales are discussed. With respect to the non-tidal mass redistributions, global plots are shown for specific epochs and known extreme situations; the corresponding maps also include marks for all worldwide known clock laboratories. The related uncertainties are estimated, and more details on the underlying theory and computation procedures are provided in the appendix along with the essential references.

3.2. Solid Earth tides

In principle, the potential induced by Earth tides at a specific station and epoch on the rigid Earth can be determined from Newton's law of gravitation and the ephemerides of the external celestial bodies. The computations can be carried out separately for the individual two-body systems (Earth–Moon, Earth–Sun, etc) with the celestial bodies regarded as point masses. The resulting potential follows from the superposition of all results. However, nowadays the tide-generating potential is expanded in solid spherical harmonics and computed by using a tidal potential catalogue. More details can be found in the appendix.

In reality, the solid Earth reacts to the tidal forces primarily like an elastic body with deformation. According to the fundamental theory on the tidal response of an elastic Earth, the deformations and potential changes caused by Earth tides are represented by applying Love numbers according to section 3.1. The Love numbers of the dominant degree 2 approximately amount to $h_2 = 0.61$ and $k_2 = 0.30$, revealing that the effective tidal potential on an elastic Earth is attenuated by a factor γ of about 0.69, compared to the tidal potential on a rigid one. It is important to note that a positive tidal potential induces an uplift of the Earth's surface, which leads to a corresponding gravity potential change with opposite sign for a station on the (deformable) Earth's surface [39]. The degree 2 zonal tidal potential ($nm = 20$) has a non-zero mean (time average) value. This time independent potential produces a permanent deformation (according to the Love number h_{20}) and consequently a time independent contribution to the potential at the Earth's surface according to the Love number k_{20} [29]. Different concepts exist for the treatment of the permanent tide [39, 40]. For a proper combination of the static part of the gravity potential with the tidal potential for relativistic redshift computations it must be ensured that the permanent tide is handled consistently.

The effective solid Earth tide potential time series are displayed in figure 1 (parts (a)–(c)) exemplarily for the year 2014 at sites NPL, PTB and NICT, respectively. The minimum values of -2.1 m^2 s^{-2} are globally uniform, while the maximum values vary between zero (at the poles) and 4.2 m^2 s^{-2} (at the equator). Hence, the tidal range (to be understood as maximum-to-minimum difference in the following) at NICT (latitude 35.7°) is larger than at NPL (51.4°) and PTB (52.3°). The semi-diurnal and diurnal tides are the main components with amplitudes modulated by periods of 2 weeks, 1 month, etc, resulting in small daily tidal variations roughly every 14 d; for further details on tidal patterns and constituents see, e.g. [41]. Furthermore, figure 1 shows corresponding differences of the tidal potential series for sites NPL–PTB (part (d)) and PTB–NICT for the entire year (part (e)) and a 4 d excerpt (part (f)); the difference signals show a range (definition as above) from 1.3 m^2 s^{-2} ($1 \cdot 10^{-17}$) for NPL–PTB to 10.0 m^2 s^{-2} ($1 \cdot 10^{-16}$) for PTB–NICT. Especially figure 1 (part (f)) shows the effect of solid Earth tides for clock comparisons over

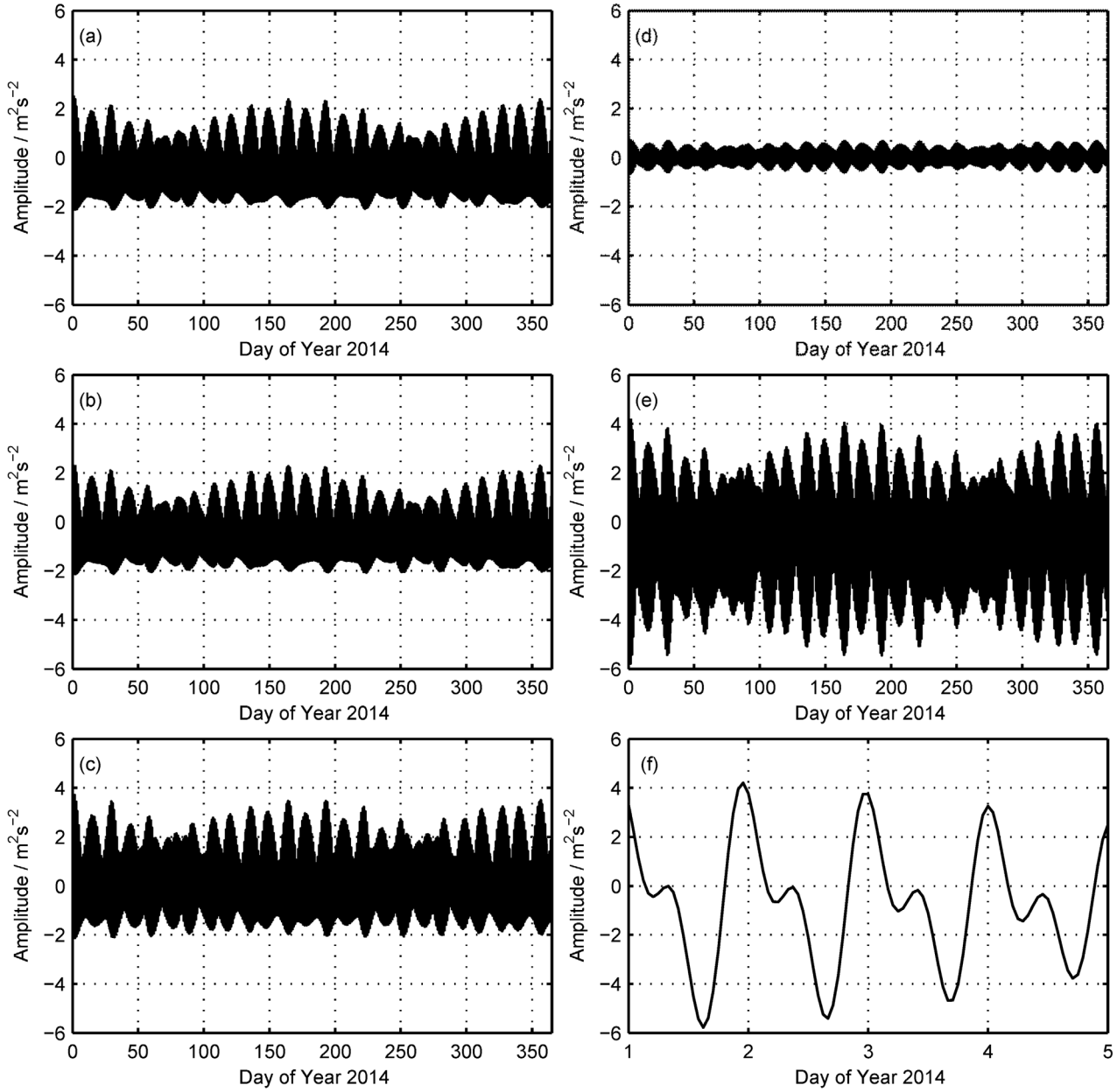


Figure 1. Effective tidal potential due to solid Earth tides for 2014 at sites NPL (a), PTB (b), and NICT (c), as well as corresponding differences between NPL–PTB (d) and PTB–NICT for the entire year (e) and a 4 d excerpt (f).

large distances with relatively short averaging times (e.g. 1 h or less), as the dominant diurnal and semi-diurnal tidal wave contributions may accumulate to a range of about 10.0 $m^2\ s^{-2}$ ($1 \cdot 10^{-16}$) within about 6 h.

The mean values of the tidal potential time series amount to $-0.6\ m^2\ s^{-2}$ for NPL and PTB as well as $-0.01\ m^2\ s^{-2}$ for NICT. This reflects the latitude dependent permanent tidal effect W_2 from [39] multiplied by the Love number combination $(1 + k_{20} - h_{20})$, and hence the tidal potential values shown at this point include the permanent deformation part, which has to be considered in conjunction with the static gravity potential values [29].

The dominant contribution to the uncertainty budget comes from the choice of the elastic Earth model for the Love numbers. However, different Earth models lead to differences at the level of only $5 \cdot 10^{-3}\ m^2\ s^{-2}$ in terms of the tidal potential.

3.3. Ocean tides

The induced signals from the ocean tidal forces (ocean tides) are called ocean tidal loading (OTL). The effective OTL potential is the summed effect of the direct attraction of the moving water masses, the elastic deformation of the Earth's crust due to the load of the water and an additional change of the gravitational potential due to the deformation [38, 42, 43]. The distribution of the continents and the topography of the ocean bottom prevent a uniform response of the ocean to the ocean tidal forces. Therefore, the tidal loading signal is generally not in phase with the solid Earth tides. Loading effects are pronounced in the diurnal and semi-diurnal waves as well as close to the coast and strongly reduce towards the continent's interior. The computation of OTL effects follows the fundamental theory developed by [37], which describes the

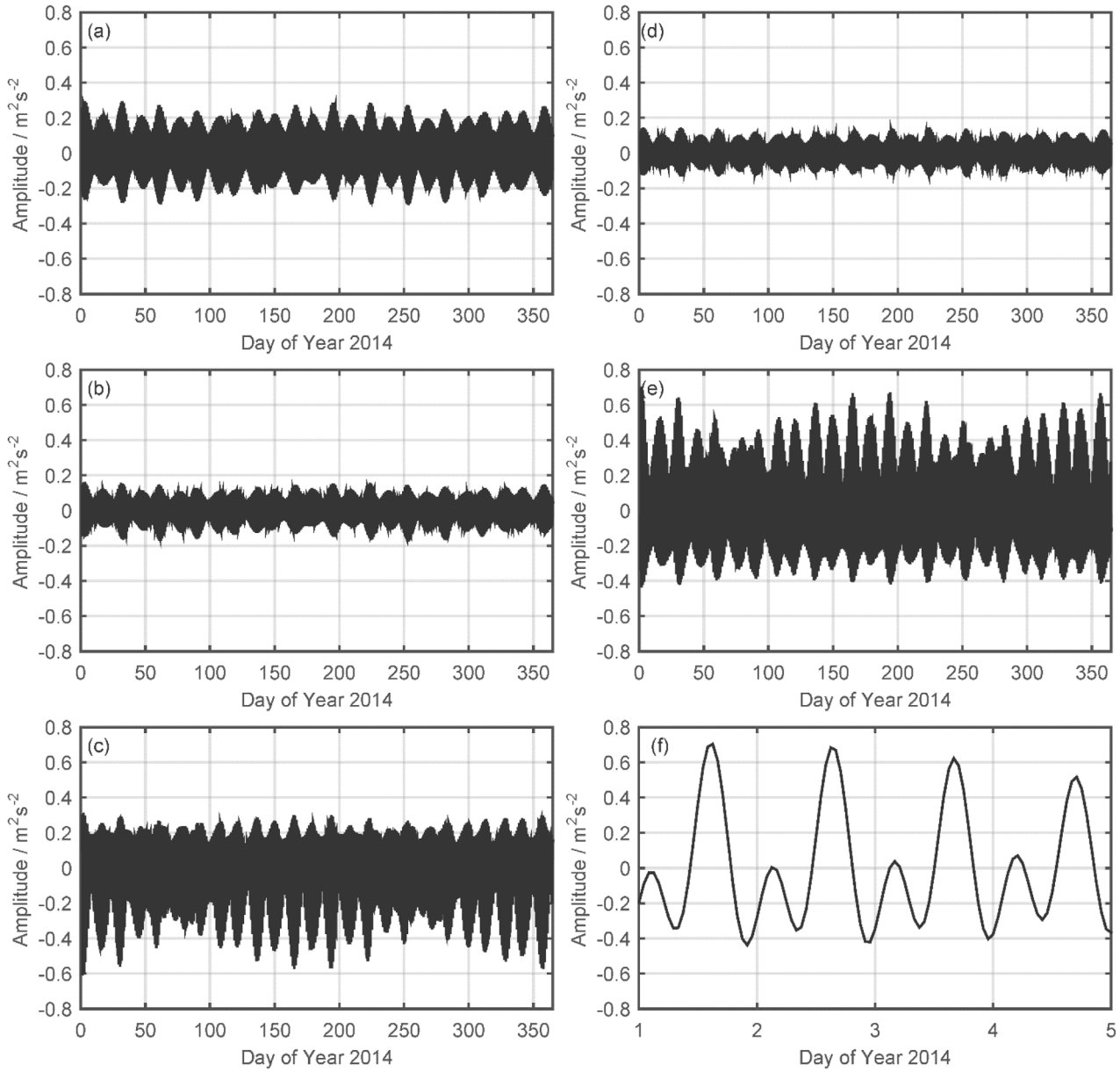


Figure 2. Effective OTL potential for 2014 at sites NPL (a), PTB (b), and NICT (c), as well as corresponding differences between NPL–PTB (d) and PTB–NICT for the entire year (e) and a 4 d excerpt (f).

response of an elastic Earth model to a point load on its surface; more details can be found in the appendix. An increase of ocean water masses (high tide) induces a positive OTL gravitational potential and a subsidence of the Earth's surface due to the load of the additional masses.

In figure 2 (parts (a)–(c)), the effective OTL potential values are displayed for the year 2014 at sites NPL, PTB and NICT, respectively. While the maximum tidal range is at the level of 0.3 $m^2\ s^{-2}$ for PTB, it doubles for NPL and triples for NICT. This corresponds to about 6, 12 and 15% of the solid Earth tides, respectively, showing the non-uniform distribution of the OTL potential and the strong dependence on the distance to the coast. Global maximum ocean tides are found at the French Atlantic coast (Bretagne) with a tidal range of 2.5 $m^2\ s^{-2}$ (50% of the solid Earth tides), which is in good agreement with the rough estimation in [21] of $2 \cdot 10^{-17}$ in terms of frequency shift (corresponds to 2 $m^2\ s^{-2}$ tidal range). Corresponding OTL potential differences are also shown in figure 2 (parts (d) and (e)); the tidal ranges amount to 0.3 $m^2\ s^{-2}$ ($3 \cdot 10^{-18}$) between NPL and PTB (part (d)) and 1.1 $m^2\ s^{-2}$ ($1 \cdot 10^{-17}$) between PTB and NICT (parts (e) and (f)). While Europe's Atlantic coasts experience similar semi-diurnal tides with two nearly equal high and low tides each day, Japan's Pacific coast shows phase shifted mixed tides with two unequal amplitudes, resulting in mainly one high and one low tide every day (see figure 2, part (f)). In any case, the maximum potential difference can arise within about 6 h.

With regard to optical clock comparisons and the definition of international time scales, a total uncertainty budget of about 2–3% (maximum values) of the OTL potential can be estimated. This corresponds to 0.02–0.03 $m^2\ s^{-2}$ at NICT and 0.01–0.02 $m^2\ s^{-2}$ (10^{-19}) at NPL in terms of gravitational

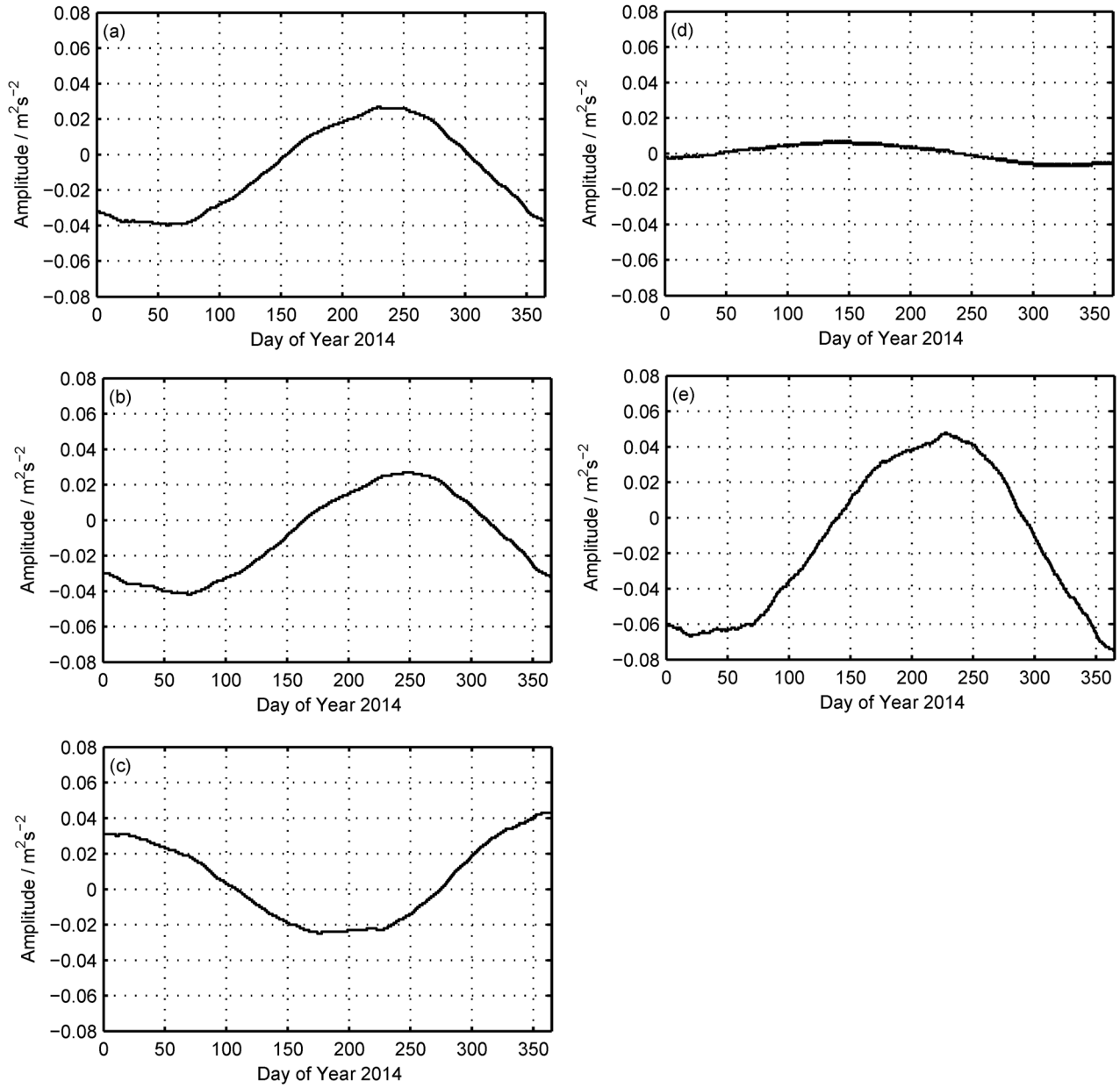


Figure 3. Effective solid Earth pole tide potential for 2014 at sites NPL (a), PTB (b), and NICT (c), as well as corresponding differences between NPL–PTB (d) and PTB–NICT (e).

potential values, and significantly better uncertainties can be expected for tidally less affected sites far away from the coast.

3.4. Solid Earth pole tides and LOD tides

Temporal changes of the Earth's rotation vector with respect to the Earth's body are described by the Earth's rotation parameters, i.e. the pole coordinates and Earth rotation angle, also known as LOD. The changes of the Earth rotation parameters mirror the combined effects of terrestrial mass redistribution and related motions. The pole tides are generated by the centrifugal effect of polar motion. It is a tide-like motion, which, however, has nothing to do with lunisolar forcing. Polar motion is almost completely described by two harmonic variations of the location of the instantaneous rotation pole with respect to the mean rotation pole of the Earth. This is an elliptical motion with an annual period (365 d), and an almost circular motion with a period of 14 months (433 d) referred to as the Chandler wobble, with the amplitudes varying in time [42]. The computation of the centrifugal potential due to polar motion and LOD on the solid Earth is described in the appendix.

In figure 3 (parts (a)–(c)), the effective solid Earth pole tide potential values are shown again for the year 2014 at sites NPL, PTB and NICT, respectively. The annual and the Chandler periods are clearly visible. The pole tide amplitudes reach extreme values of 0.04 $m^2\ s^{-2}$ at latitudes of $\pm 45°$ with decreasing values towards the equator and the poles. Consequently, the amplitudes at NPL and PTB are slightly larger than those at NICT with corresponding ranges of 0.07 $m^2\ s^{-2}$ and 0.05 $m^2\ s^{-2}$, respectively. The pole tide potential differences between laboratory sites reach maximum

values for longitude differences of 180°; for PTB–NICT (figure 3, part (e)), the range is 0.12 $m^2\ s^{-2}$ ($1 \cdot 10^{-18}$), whereas for NPL–PTB (figure 3, part (d)) the range remains at a level of 0.01 $m^2\ s^{-2}$ ($1 \cdot 10^{-19}$). The uncertainties of the pole coordinates provided by the IERS are nowadays at the level of only 10^{-10} rad and thus do not contribute significantly to the uncertainty budget. The effective LOD tide potential is not shown here as the maximum amplitudes are only 0.004 $m^2\ s^{-2}$ at the equator and 0.001 $m^2\ s^{-2}$ ($1 \cdot 10^{-20}$) at the selected laboratory sites.

3.5. Ocean pole tides

The ocean pole tides are generated by the centrifugal effect of polar motion on the oceans. For the long periods of polar motion (annual and 14 months), the ocean pole tides are expected to have an equilibrium response, where the displaced ocean surface is in equilibrium with the forcing equipotential surface [29, 44] presents a self-consistent equilibrium model for the ocean pole tides, with the ocean pole tides producing perturbations to the normalized spherical harmonic coefficients of the potential up to degree and order 360, yielding the total potential according to terms $(1 + k'_n)$. In addition, the ocean pole tides produce deformations at any point on the surface of the Earth. In [29], the transformation between the ocean pole load tide coefficients and vertical displacements according to the load Love numbers h'_n is given for individual cells of a global 0.5° × 0.5° grid. The contributions of the total potential and the additional potential due to vertical displacements have the same signs and consequently add up for the effective potential.

Approximately 90% of the variance of the ocean pole tide is provided by the degree 2 components with dominant coefficients of $nm = 21$. Utilizing the pole coordinates from the year 2014, the effective ocean pole tide potential up to degree and order 2 shows maximum amplitudes of 0.01 $m^2\ s^{-2}$, occurring at latitudes of ±45°. The range of the effective potential at the laboratory sites reaches 0.015 $m^2\ s^{-2}$, while the differences between PTB and NICT have a range of 0.03 $m^2\ s^{-2}$ ($3 \cdot 10^{-19}$), and thus the effects over continental scales are fully negligible (below 0.001 $m^2\ s^{-2}$).

3.6. Atmospheric tides

The atmosphere undergoes tidal effects caused by solar heating (atmospheric tides). In practice, two constituents may be of relevance, i.e. the solar diurnal and semi-diurnal tidal waves S1 and S2, respectively. The major part of this signal is not a gravitational effect, but results from the diurnal and semi-diurnal temperature variations (thermic effect), inducing an atmospheric pressure loading on the Earth's surface. The effects of the atmospheric tides can be computed from a tidal model, e.g. by [45], as applied by GFZ, or by [46], being the conventional model of [29]. Both models are derived from global surface pressure fields from the European Centre for Medium-Range Weather Forecasts (ECMWF).

Global grids of the amplitudes and phases of vertical surface displacements due to the S1 and S2 waves are provided in [29], based on the model from [46], and these quantities can then be converted to potential values. Both the S1 and S2 amplitudes are at the level of 0.0015 m and most pronounced in the tropics around the equator, where no laboratories operating or building optical clocks exist so far. At mid-latitudes, the vertical displacements reduce to the level of about $5 \cdot 10^{-4}$ m, leading to potential changes with amplitudes of about 0.005 $m^2\ s^{-2}$ ($5 \cdot 10^{-20}$) and twice this amount for the difference signal range between laboratories, which is fully negligible for optical clock comparisons and the definition of international time scales, although it is regularly considered in the GRACE atmosphere and ocean de-aliasing products (see next section).

3.7. Non-tidal mass redistributions in the atmosphere

Non-tidal mass redistributions due to atmospheric pressure changes induce variations of the gravity potential, which are pronounced at short time scales (from hours to days), but also occur at seasonal scales. Strong short-term variations are related to (anti)cyclones from local to continental scales with amplitudes up to ±60 hPa, while seasonal variations are much smaller with amplitudes up to ±10 hPa [47]. Potential effects at stations on the Earth's surface are caused by the direct attraction of the air masses (attraction or direct effect) and by the vertical displacement of the Earth's surface, inducing an additional potential change due to the redistribution of the masses (elastic deformation or indirect effect). For a thorough introduction of atmospheric effects in geodesy see [48]. Nowadays, the computation of non-tidal atmospheric effects is almost exclusively based on 3D atmospheric pressure data, e.g. from the ECMWF. The computation of the total potential values, vertical displacements and the effective potential based on spherical harmonic expansions is described in the appendix for the non-tidal mass redistributions considered in sections 3.7–3.9, along with an introduction of the applied models and products.

Figure 4 shows the total potential according to terms $(1 + k'_n)$ (part (a)) and the vertical displacements due to atmospheric pressure variations according to the load Love numbers h'_n (part (b)), utilizing the spherical harmonic coefficients up to degree and order 100 (spatial resolution of 200 km, 6 h intervals) from the ATM data set of the GRACE Atmosphere and Ocean Level-1B De-Aliasing Product (AOD1B [49, 50]). The variations represent deviations from a long-term average of the 3D atmospheric mass distribution as a certain reference state. At the given date of 6 UTC on January 30, 2000, a typical storm surge occurred in the North Sea associated with low atmospheric pressure [51]. The induced positive total potential values are clearly visible over Central and Northern Europe, resulting from a decrease of the atmospheric masses above the surface of the Earth. A mass decrease above the Earth's surface in turn leads to an uplift of the surface, i.e. a positive vertical displacement, as a consequence of the reduced load over Central and Northern Europe. Simultaneously, high atmospheric pressure with induced negative potential and vertical displacement values occur over large parts of Asia and North America, related to the more seasonally pronounced

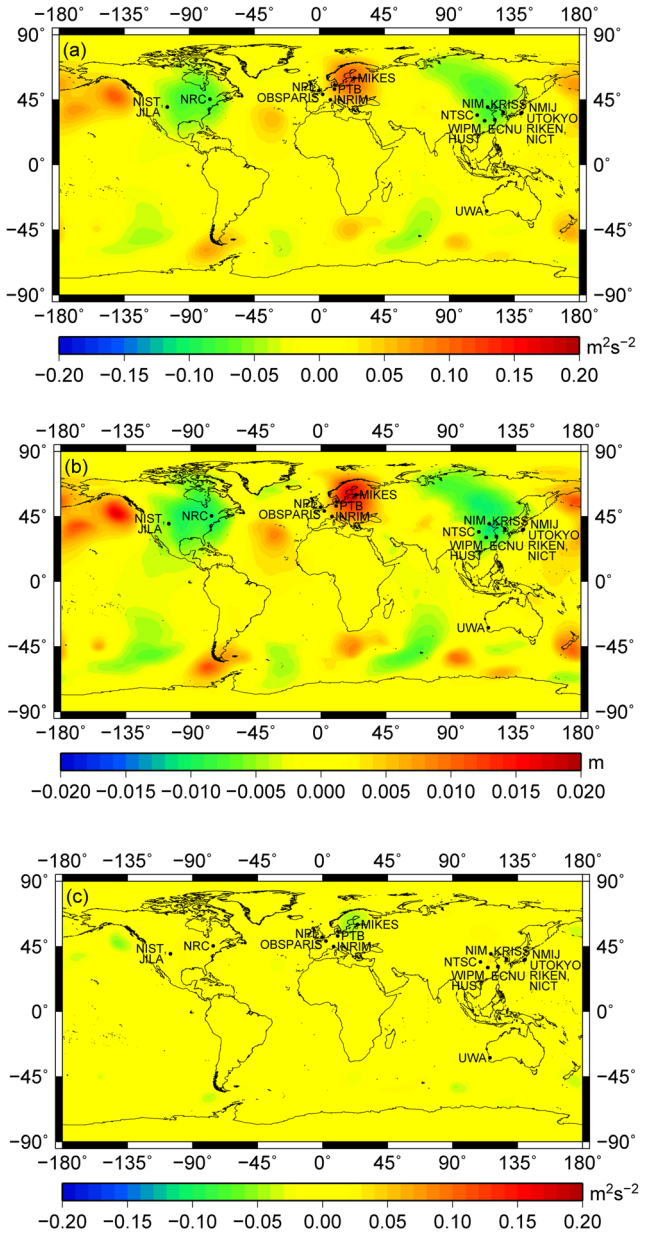


Figure 4. Total gravitational potential according to terms $(1 + k'_n)$ (a), vertical displacement of the Earth's surface (b), and effective gravitational potential according to terms $(1 + k'_n - h'_n)$ (c), all related to atmospheric pressure variations at 6 UTC on 30 January 2000.

Siberian High (Siberian anticyclone) and the kindred North American High (Canadian anticyclone). The maximum amplitudes at the laboratory sites are at the level of ± 0.10 m^2 s^{-2} and ± 0.015 m, respectively. Figure 4 (part (c)) illustrates the effective potential according to terms $(1 + k'_n - h'_n)$, which is significantly smaller, as the total potential and the effects of the vertical displacements cancel out to a large extent. The effective potential reaches amplitudes of only a few 0.01 up to 0.05 m^2 s^{-2} $(5 \cdot 10^{-19})$ at as well as between the laboratory sites. The maximum impact of the AOD1B product uncertainties on the induced gravitational potential is estimated to 0.005 m^2 s^{-2} [52].

3.8. Non-tidal mass redistributions in the oceans

Non-tidal mass redistributions in the oceans are mostly caused by atmospheric pressure variations and strong winds. The effective gravity potential again consists of the direct attraction of the water masses and the vertical displacement of the Earth's surface as well as additional potential changes due to the deformation of the Earth's crust. Non-tidal ocean loading is most significant near the coast, where the variation in ocean bottom pressure is large. Vertical displacements from non-tidal ocean loading are usually at the level of 0.002–0.003 m seasonally, but can reach up to 0.02 m (range) within hours and days, e.g. due to storm surges [53].

[51] compute non-tidal ocean loading effects for a typical storm surge in the Southern North Sea at 3 UTC on January 30, 2000, with an increased non-tidal sea-level of over 2 m. The induced vertical displacements amount up to 0.03 m right at the coast. At 6 UTC on January 30, 2000, the total potential and the vertical displacements from the OCN data set of AOD1B [49, 50], based on ocean bottom pressure variations from the OMCT (Ocean Model for Circulation and Tides [54]), are displayed in figure 5 (parts (a) and (b)), respectively. The variations represent deviations from a long-term average of the ocean bottom pressure. The total potential in Central and Northern Europe increases due to the increased water masses up to the level of 0.1 m^2 s^{-2} at the coast but is already below 0.05 m^2 s^{-2} at NPL. Simultaneously, the inverse conditions occur in East Asia and North America. The induced vertical displacements in Europe go down to -0.01 m at NPL and -0.005 m at PTB, while they amount up to $+0.005$ m at NICT; these figures are similar to those provided by [51]. For the effective potential relevant for the optical clock comparisons and the definition of international time scales (part (c)), both effects add up, reaching amplitudes of more than 0.1 m^2 s^{-2} at NPL and NICT. Consequently, the potential differences reach the level of more than 0.2 m^2 s^{-2} $(2 \cdot 10^{-18})$ over intercontinental scales (e.g. NPL–NICT), but also reach 0.1 m^2 s^{-2} $(1 \cdot 10^{-18})$ over continental scales (e.g. NPL–INRIM), with possibly twice these values for the corresponding ranges. The maximum impact of the AOD1B product uncertainties on the induced gravitational potential is estimated to 0.005 m^2 s^{-2} [52].

3.9. Non-tidal hydrological mass redistributions

Variations in the continental water storage comprise storage changes in snow, soil moisture, groundwater and surface water. The effective gravity potential consist of the direct attraction of the water masses on or below the surface of the Earth and the vertical deformation of the Earth's crust inducing additional potential changes. The detection of continental water storage changes from geodetic measurements requires the prior reduction of atmospheric and oceanic effects (see previous sections). For the determination of large- and medium-scale gravity field variations due to seasonal and long-term continental water storage changes, monthly solutions from the GRACE mission or global hydrological models such as

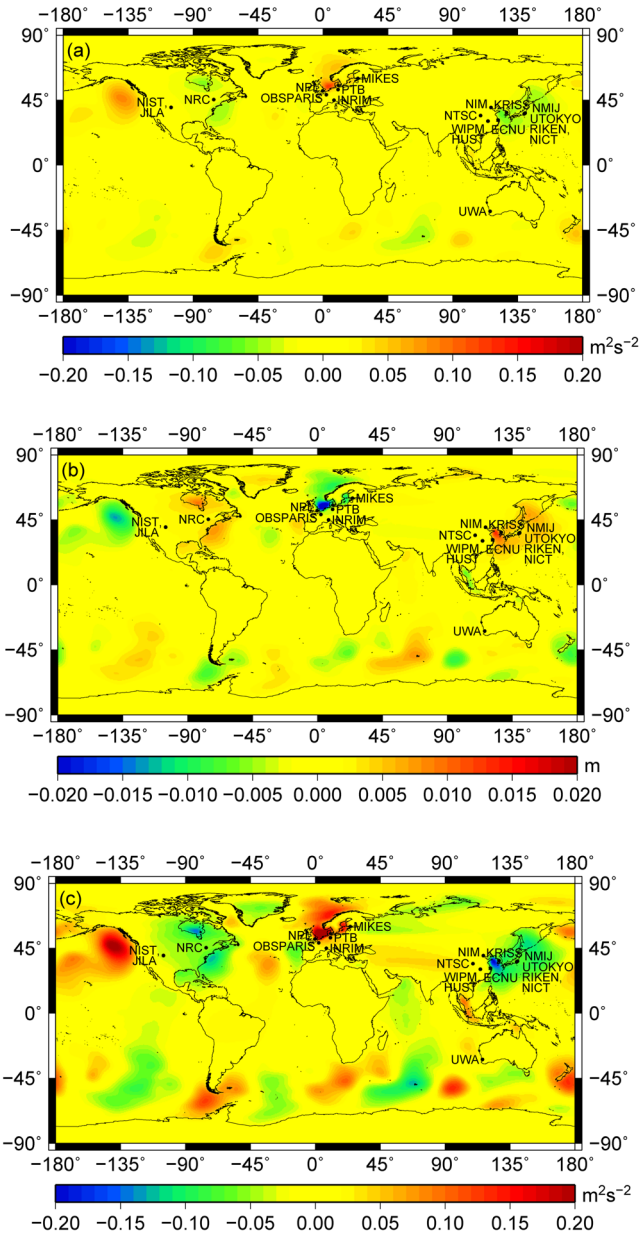


Figure 5. Total gravitational potential according to terms $(1 + k'_n)$ (a), vertical displacement of the Earth's surface (b), and effective gravitational potential according to terms $(1 + k'_n - h'_n)$ (c), all related to non-tidal ocean loading at 6 UTC on 30 January 2000.

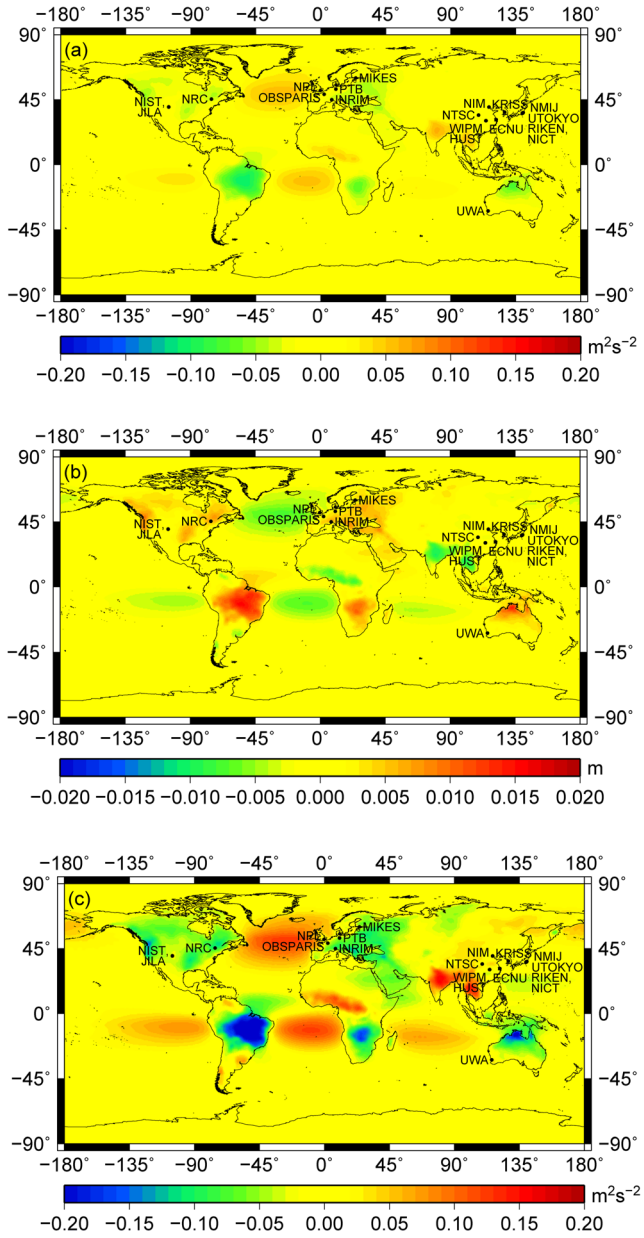


Figure 6. Differences due to continental water storage from GLDAS between September 2003 and March 2004 for the total gravitational potential according to terms $(1 + k'_n)$ (a), the vertical displacement of the Earth's surface (b), and the effective gravitational potential according to terms $(1 + k'_n - h'_n)$ (c).

NASA GLDAS (Global Land Data Assimilation [55]) can be employed [56].

The signals due to continental water storage changes are pronounced on seasonal scales. In Central Europe, typical seasonal variations reflect gravity potential maxima and vertical displacement minima in winter to spring (wet, snowy) and vice versa in summer to autumn (mostly dry). Apart from these seasonal variations, hydrological loading signals can be associated with major precipitation events, e.g. heavy rainfalls and induced floods. For the estimation of seasonal effects induced by continental water storage changes, monthly spherical harmonic coefficients up to d/o 100 based on GLDAS are used. The differences of the total potential and vertical displacement induced by continental water storage between September 2003 and March 2004 are shown in figure 6 (parts (a) and (b)), respectively. These can be associated with the 2003 summer heat wave in Europe. The seasonal range of the total potential remains within 0.05 $m^2\ s^{-2}$, while the vertical displacements reach the level of 0.005 m at the laboratory sites. A decrease of the continental water storage induces a decrease of the total potential and an uplift of the Earth's surface due to the decrease of the hydrological load. Hence, the terms of the effective potential in figure 6 (part (c)) add up to ranges of 0.1 $m^2\ s^{-2}$ at the laboratory sites and differences of up to 0.1 $m^2\ s^{-2}$ over continental scales and 0.2 $m^2\ s^{-2}$ over intercontinental scales. The largest seasonal signal, however, occurs in the Amazon River basin with a maximum amplitude range

of 0.3 $m^2\ s^{-2}$. Non-seasonal variations, e.g. due to floods, usually do not exceed the level of a few 0.01 $m^2\ s^{-2}$ at the laboratory sites. The uncertainty of the GLDAS loading products are estimated to be at the level of better than 0.001 m for the vertical displacements [57], corresponding to 0.01 $m^2\ s^{-2}$ in terms of potential.

The effects of continental water storage are superimposed by local hydrological mass redistributions resulting from site-specific variations of local snow, soil moisture, groundwater and surface water up to a distance of some 100 m around a specific station. It is well known that these occur mainly seasonally, but that short-term fluctuations of a few hours to a few days can contribute significantly, e.g. due to melting snow and resulting water movements [58]. Local hydrological gravity effects are inherent in superconducting gravimeter observations [59]. For a rough estimation, maximum local hydrological gravity variations from an area of 1 km around the station can be assumed with $100 \cdot 10^{-9}$ m s^{-2}. The maximum local hydrological potential variations follow from the contribution of an inner zone by applying Stokes formula [42], amounting to $5 \cdot 10^{-5}\ m^2\ s^{-2}$ only. The estimation of the vertical displacement due to local hydrological mass redistributions can be inferred from [60], being insensitive at the level of up to 0.001 m within distances of 120 km, but captured by the hydrological loading grids based on global hydrological models. It can be stated that the local hydrology does not play a role in the context of optical clock comparisons and the definition of international time scales.

3.10. Sea-level variations

Time variations of the sea level are relevant for the definition of the zero level or height reference surface (with the gravity potential W_0 [29]). Historically, tide gauge data have been used for the definition and realization of national vertical reference systems. Typically, countries have chosen one or more tide gauge stations and then averaged over some time period in order to determine mean sea level as the vertical datum, which was believed to be constant at the very beginning. However, the mean sea level today usually does not agree with the local vertical datum due to relative sea level rise and fall [61], e.g. the mean sea level at Marseille, France, is approx. 0.10 m above the local datum from 1885–1897, whereas it is approx. 0.15 m above the Ordnance Datum (1915–1921) at Newlyn, UK (status of the year 2000). Tide gauge stations of the European vertical reference network (EUVN) provide different (relative) trends of the annual mean sea level rise and fall of a few 0.001 m a^{-1} (0.01 $m^2\ s^{-2}\ a^{-1}$), without a clear regional pattern, except for the Baltic Sea, where the glacial isostatic adjustment (GIA) (see next section) dominates the tide gauge signals [61]. Global maximum values of sea level rise, however, reach a level up to 0.01 m a^{-1} (0.1 $m^2\ s^{-2}\ a^{-1}$) in the Pacific Ocean, e.g. in the Philippine Sea.

Additionally, the mass redistributions in the oceans again lead to the deformation of the Earth's crust in coastal regions, also known as vertical land movement (VLM), and are observed by GNSS techniques in order to separate the secular

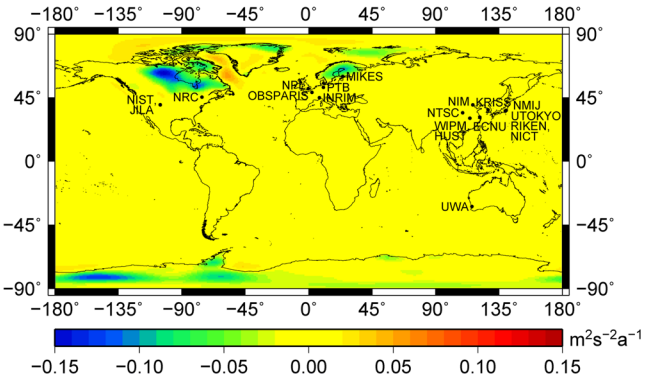


Figure 7. Rates of the effective gravitational potential according to terms $(1 + k'_n - h'_n)$ due to GIA from the ICE-5G (VM2) model.

climate related components of changes in mean sea level from VLM. However, in many cases, observations cannot assure statistically significant results due to the present vertical GNSS accuracy, which should be improved by the extension of the time series into the future [62]. Issues on the long-term stability of terrestrial reference systems at global and continental scales, taking into account different time-dependent influences of the system Earth, can be found, e.g. in [63].

3.11. Solid Earth processes

Time-variable gravity potential changes driven by solid Earth processes are very manifold in terms of temporal and spatial scales. Again, these are the combined effects of gravity field changes (total potential) and of the vertical displacement of stations on the surface of the Earth. Both are inherent in geometric levelling. Instead, GNSS measurements, nowadays the primary tool for investigations of recent crustal deformation, are not sensitive to mass redistributions, and the corresponding gravity field effect has to be determined separately. However, the latter effect remains usually below 10% of the height change and can often be neglected. Several examples of solid Earth processes, inducing gravity field variations and crustal deformations, are given in [42].

One of the main contributors is the postglacial rebound, which is a GIA process. This is the response of the Earth to the changing surface ice load at the recent deglaciation cycle, which started 21 000 years ago. It is well understood and observable with modern geodetic techniques, such as GNSS, absolute gravimetry, but also with GRACE. The impact of GIA is well pronounced in polar regions, but can also affect regions of mid-latitude to a certain extent. GIA effects in terms of the total potential and vertical displacements are, e.g. provided by the ICE-5G (VM2) model [64]. The total trend for the potential goes up to 0.01 $m^2\ s^{-2}\ a^{-1}$ in Canada and about half this value in Fennoscandia. The maximum land uplift rate, however, is much more pronounced with up to 0.017 m a^{-1} and 0.008 m a^{-1} in Canada and Fennoscandia, respectively. Consequently, the rates of the effective potential due to GIA are $-0.15\ m^2\ s^{-2}\ a^{-1}$ in Canada and half this value in Fennoscandia (figure 7). This is generally taken into account in national height systems of the Northern countries, e.g. in Sweden [65]. However, the clock laboratory sites are hardly

Table 1. Global (absolute) maximum amplitudes in $m^2\ s^{-2}$ of time-variable gravity potential components and corresponding locations along with dominant time periods.

Phenomenon	Effect	Maximum amplitude	Location	Dominant time periods
Tides	Solid Earth tides	4.2	Equator	From permanent to semi-diurnal
	Ocean tides	1.25	French Atlantic coast	Diurnal and semidiurnal
	Solid Earth pole tides	0.04	Latitudes of ±45°	Annual and 433 d
	LOD tides	0.004	Equator	Secular
	Ocean pole tides	0.01	Latitudes of ±45°	Annual and 433 d
	Atmospheric tides	0.03	Tropics	Diurnal and semidiurnal
Non-tidal mass redistributions	Atmosphere	0.05	Global	Hours and days
	Oceans	0.2	Bights	Hours and days
	Continental water storage	0.15	Amazon River basin	Seasonal
	Local hydrology	—	—	Seasonal
	Sea level variations	0.1/a	Pacific Ocean	Secular
	Solid Earth processes (GIA)	0.15/a	Canada	Secular

Table 2. Range (maximum-to-minimum) in $m^2\ s^{-2}$ of time-variable gravity potential components at laboratory sites and differences between laboratory sites over continental (Europe) and intercontinental (Europe–Japan) scales.

		Range (maximum-to-minimum)		
Phenomenon	Effect	At laboratory sites	Over continental scales	Over intercontinental scales
Tides	Solid Earth tides	5.8	1.3	10.0
	Ocean tides	0.9	0.3	1.1
	Solid Earth pole tides	0.07	0.01	0.12
	LOD tides	0.001	0.001	0.001
	Ocean pole tides	0.015	0.001	0.03
	Atmospheric tides	0.01	0.001	0.02
Non-tidal mass redistributions	Atmosphere	0.05	0.05	0.1
	Oceans	0.2	0.2	0.4
	Continental water storage	0.1	0.1	0.2
	Local hydrology	—	—	—
	Sea level variations	0.01/a	0.01/a	0.01/a
	Solid Earth processes (GIA)	0.05/a	0.05/a	0.05/a

affected, except NRC (National Research Council of Canada) and MIKES (Mittatekniikan keskus, Finland) with maximum rates of $-0.05\ m^2\ s^{-2}\ a^{-1}$.

While horizontal velocities derived from the international terrestrial reference network (ITRF) are mainly caused by the motion of the tectonic plates, the detection of vertical motions on the global scale is still problematic. However, large-scale crustal deformations at diverging tectonic plate boundaries or in collision zones can be observed by GNSS measurements in continuously operating regional or local networks. Vertical displacements due to short term co-seismic effects can reach the level of several meters close to the epicentre of major earthquakes and a few 0.01 m up to distances of some 100–1000 km (depending on the earthquakes magnitude). These are superimposed by long-term post-seismic effects with much smaller amplitudes [42]. Volcanic activities are more pronounced on local scales. While the laboratory sites in Europe are hardly affected by major earthquakes and volcanic activities, this does not hold for the laboratory sites in North America and East Asia. Particularly Japan is affected, being part of the 'Ring of Fire'.

Other large-scale vertical displacements have been found, e.g. in the European Alps, with mountain uplift rates of $0.001\ m\ a^{-1}$ and more, probably caused by erosional unloading, but maybe also due to tectonic processes. Man-made vertical crustal movements are generally of more local character and related to natural gas, oil and geothermal fields, the withdrawal of groundwater, mining and load changes in water reservoirs, among others [42]. However, in general, these effects do not have to be considered with regard to an accuracy level of 10^{-18}.

4. Summary and conclusions

Time-variable gravity potential components induced by tides and non-tidal mass redistributions were investigated with regard to optical clock comparisons and the definition of international time scales. Rough estimates of the amplitudes by [21] could be confirmed by applying state-of-the-art models, data sets and software, widely used in high precision geodesy. The investigated components of the time-variable gravity potential are summarized in tables 1 and 2. The global maximum amplitudes and the dominant time periods for all investigated individual gravity potential components are given in table 1, while table 2 provides the signal ranges

(maximum-to-minimum) at selected laboratory sites and for corresponding differences between laboratory sites over continental and intercontinental. Thus, table 1 is mainly related to contributions to international time scales, as it provides the maximum amplitudes along with the corresponding locations of all investigated effects, which are needed to refer the clocks to a conventional zero level surface W_0, while, on the other hand, table 2 provides the corresponding signal ranges, which are of main interest for clock comparisons.

By far the most dominant component is the potential induced by solid Earth tides, which has to be taken into account for the definition of international time scales and optical clock comparisons over intercontinental scales below the 10^{-16} level, while clock comparisons over continental scales are affected only at the 10^{-17} level. The uncertainty of this component has a negligible level of $5 \cdot 10^{-20}$, with the dominant contribution resulting from the choice of the elastic Earth model for the determination of the Love numbers. The induced signals of ocean tides, being the second largest contribution, should be taken into account for clock comparisons over intercontinental scales and the definition of time scales at the 10^{-17} level, but also for comparisons over continental scales at the 10^{-18} level. The maximum uncertainty is estimated to be a few parts in 10^{-19}, but significantly better uncertainties can be expected for tidally less affected sites far away from the sea. Hence, the solid Earth and ocean tides are the only two contributors, being relevant for clocks with a performance at the level of 10^{-17}.

However, several other time-variable effects can affect clock performances at the 10^{-18} level, in particular over intercontinental scales. The well-known solid Earth pole tides reach the 10^{-18} level at laboratory sites and between sites over intercontinental scales, while they are negligible over shorter distances. It was also shown that non-tidal mass redistributions in the atmosphere, the oceans and the continental water storage can reach the 10^{-18} level (with an uncertainty of 10^{-19}) under exceptional weather conditions, both at laboratory sites and between sites over intercontinental scales, while they reduce significantly over shorter distances and under normal weather conditions. Furthermore, secular signals due to sea level changes and GIA exist, which are important in particular for the realization of international time scales.

State-of-the-art models, data sets and software were applied in the framework of this study. However, so far no tailored products or services exist for the computation of time-variable gravity potential components, as needed by the optical clock community or for international time scales. The key contribution from geodesy and geophysics in this context, remaining a future task, will be the development of operational tools and products for a consistent reduction of all tidal and non-tidal effects, as applied, e.g. in the context of GRACE, GNSS and superconducting gravimetry. Furthermore, this task is also an essential prerequisite for the determination of static potential differences with optical clocks in the framework of chronometric levelling at the level of 10^{-18}.

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Appendix

A.1. Solid Earth tides

The tide-generating potential is computed by using a tidal potential catalogue, i.e. a table of amplitudes, phases and frequencies for a specific number of tidal waves. Tidal potential catalogues consist of the time dependent potential coefficients C_i^{nm}, S_i^{nm} of degree n and order m for each tidal wave i, given in units of $\mathrm{m^2\,s^{-2}}$, and integer coefficients for the computation of the time dependent arguments α_i. The tidal potential on a rigid Earth at a specific point with geocentric spherical coordinates $(\theta,\ \lambda,\ r)$ is represented by

$$V^{\mathrm{rigid}}(t) = \sum_{n=2}^{n_{\max}} \sum_{m=0}^{n} \left(\frac{r}{a}\right)^n \overline{P}_{nm}(\cos\theta) \times \sum_{i=1}^{i_{\max}} \{C_i^{nm}(t)\cos[\alpha_i(t)] + S_i^{nm}(t)\sin[\alpha_i(t)]\}, \tag{A.1}$$

where a is the semi-major axis of the Earth (geodetic reference system 1980), t is the time from the reference epoch J2000 in Julian centuries, $\overline{P}_{nm}(\cos\theta)$ are the fully normalized associated Legendre functions of degree n and order m [67]. For the computation of the tide-generating potential, the tidal potential catalogue HW95 [68] is used. The harmonic development of HW95 contains 12 935 tidal waves, which are attributed to 54 tidal groups, denoted by the main tidal wave of the specific group, e.g. M2 for the lunar semidiurnal wave. Besides a significant permanent tide, the periods of the tidal waves go down to 4 h and up to several years with most dominant diurnal and semidiurnal contributions. The expansion is carried out up to a maximum degree $n_{\max} = 6$.

The effective tidal potential on the solid Earth for a specific point on the deformable surface of the Earth is then

$$V^{\text{elastic}}(t) = \sum_{n=2}^{n_{\max}} \sum_{m=0}^{n} (1 + k_{nm} - h_{nm}) \left(\frac{r}{a}\right)^n \overline{P}_{nm}(\cos\theta) \times \sum_{i=1}^{i_{\max}} \{C_i^{nm}(t)\cos[\alpha_i(t)] + S_i^{nm}(t)\sin[\alpha_i(t)]\}. \quad \text{(A.2)}$$

The Love numbers k_{nm}, h_{nm} are computed for the geophysical Earth model PREM (preliminary Earth reference model [69]) with slight surface layer modifications, i.e. an elliptical, uniformly rotating, oceanless Earth with liquid outer core and inelastic mantle according to the Wahr–Dehant–Zschau model [70].

For the computation of the tidal potential (A.2) for stations on the deformable surface of the Earth, the program PREDICT from the Earth tide data processing package ETERNA 3.4 [71] is chosen and modified accordingly. The original version of ETERNA 3.4 can be downloaded, e.g. from the websites of the GGP (global geodynamics project, www.eas.slu.edu/GGP/ETERNA34/). For the modified version of PREDICT please contact the authors. PREDICT provides the values as reductions, which have to be subtracted from the static gravity potential in (5).

With regard to the uncertainty budget of predicted tidal potential time series, the HW95 tidal potential catalogue is supposed to be the most accurate tidal development with an uncertainty better than $0.01 \cdot 10^{-9}$ m s^{-2} RMS in terms of tidal acceleration [68, 72]. With a typical tidal range of $2200 \cdot 10^{-9}$ m s^{-2} for the tidal acceleration, this corresponds to $5 \cdot 10^{-6}$ of the tidal range and thus amounts to only $2.5 \cdot 10^{-5}$ m^2 s^{-2} RMS for the tidal potential. Other tidal potential catalogues, e.g. by Tamura or Roosbeek, have a reduced accuracy by at least a factor of 10, mostly due to the reduced number of tidal waves included. However, also these catalogues could be used with regard to the present accuracy requirements for optical clocks. The dominant contribution to the uncertainty budget comes from the choice of the elastic Earth model at the level of $1.5 \cdot 10^{-3}$ of the tidal range [42, 73]. Besides the applied Wahr–Dehant–Zschau model, other models like, e.g. the Dehant–Defraigne–Wahr model could be used, either as a hydrostatic or a non-hydrostatic and inelastic model [74]. The discrepancies between these models lead to differences at the level of only $5 \cdot 10^{-3}$ m^2 s^{-2} in terms of the tidal potential.

A.2. Ocean tides

Basically, the computation of OTL consists of three steps [43]. First, the degree dependent load Love numbers h'_n, k'_n, l'_n have to be computed for a geophysical reference Earth model like PREM. However, due to their dependency on the location and the dimensions of the load, these have to be computed up to very high degrees, e.g. $n_{\max} = 10\,000$. The second step is the determination of the Green's function for the relevant functional, herein the potential at the Earth's surface, formed by weighted sums of the load Love numbers and the Legendre polynomials, depending on the spherical distance ψ between the computation point and the surface mass load. The Green's function of the effective potential at the Earth's surface is

$$G_{V'}(\psi) = \frac{ag}{M_{\text{E}}} \sum_{n=0}^{\infty} (1 + k'_n - h'_n) P_n(\cos\psi) \quad \text{(A.3)}$$

with the mean radius of a spherical Earth a, an average gravity value at the Earth's surface g, the mass of the Earth M_{E} and the Legendre polynomials $P_n(\cos\psi)$ with the specific combination of load Love numbers [38, 75]. The Green's function describes the impact of a 1 kg point-load at a spherical distance ψ on the potential at the computation point. In the final step, the OTL potential is computed by a convolution integral over the sphere (in practice over the oceans) of the Green's function, herein for the gravitational potential at the Earth's surface, and tidal heights from an ocean tide model. The OTL potential is

$$V'(\theta, \lambda) = \rho \int_0^{2\pi} \int_0^{\pi} H(\theta', \lambda') G_{V'}(\psi) \mathrm{d}S', \quad \text{(A.4)}$$

where (θ, λ) are the geocentric coordinates of the computation point, ρ is the mean density of sea water, H is the tidal height of the specific tidal wave at the point (θ', λ') provided by ocean tide models, describing the load on the differential subsurface area $\mathrm{d}S' = a^2 \sin\theta' \mathrm{d}\theta' \mathrm{d}\lambda'$ on a sphere with radius a [38, 76]. The mass distribution is represented by a constant density layer with varying tidal height covering a sphere of radius a. The area element $\mathrm{d}S'$ times the tidal height produces a volume, which results in a mass by multiplying it with the density of the sea water. The multiplication with the Green's function gives the contribution of this water mass body at (θ', λ') to the OTL potential at (θ, λ). For the prediction of the OTL potential, a large number of tidal constituents has to be used, which are found by interpolation of the few tidal waves included in the ocean tide models [38].

For the computation of the OTL potential time series at a specific station at the Earth's surface, the current version of the ocean tide loading processing package SPOTL 3.3.0.2 [77, 78] is used. This can be downloaded from the websites of the Scripps Institution of Oceanography at the University of California (http://igppweb.ucsd.edu/~agnew/Spotl/spotlmain.html). SPOTL computes the so-called potential height (gravitational potential (A.4) divided by gravity g), which has to be multiplied again by a nominal gravity value ($g = 9.81$ m s^{-2}) in order to get the effective potential relative to the deformable surface of the Earth according to [75]. At high tide, the potential heights from SPOTL are positive, while the Earth's surface is subsiding due to the load of the increased water masses. Consequently, the effects of the total potential according to terms $(1 + k'_n)$ and the potential changes due to vertical displacement add up for the effective potential values according to $(1 + k'_n - h'_n)$. These have to be added to the static gravity potential values in (5). SPOTL is the only available software to provide the effective OTL potential. The implemented Green's function is based on the Harkrider model, which is a modified version of the Gutenberg-Bullen A Earth model [37], with the top 1000 km replaced by the oceanic crust and mantle structure of [79]. The global ocean tide model EOT11A with

a grid spacing of $7.5' \times 7.5'$ is applied with the main astronomical tides (semi-diurnal and diurnal) M2, S2, N2, K2, O1, K1, P1 and Q1 as well as the non-linear constituent M4 and long-period tides Mm and Mf.

With regard to the uncertainty budget of the OTL potential from SPOTL, the discrepancies in the OTL caused by different Green's functions are regarded as primary uncertainty source [80], resulting in maximum differences at the level of 2% for the OTL amplitudes [81]. Simplifications of the physical model are primarily the assumptions of a constant value of sea water density and of a spherical Earth, which causes errors in the OTL computations of 1% and 0.3%, respectively. The maximum discrepancies between the OTL effects from various OTL computation programs under investigation in [80] are at the level of 2–5%, as already stated in [78], however based on a previous version of SPOTL. In the recent version of SPOTL 3.3.0.2, the bilinear interpolation of the ocean tide model instead of using the nearest tidal grid value was implemented as well as a variable seawater density based on a global database [77]. Thus, the applied version of SPOTL should have an uncertainty of better than 1%. The impact of different ocean tide models on the OTL gravity component is numerically tested for the NPL site. The differences show a range in the dominant M2 amplitude of 2.5%, except for TPXO7.2, which leads to maximum discrepancies of 5%, showing that the uncertainty in the ocean tide models improved drastically. However, it is still one of the limiting factors in OTL computations. At present, no single ocean tide model gives completely satisfactory results in all areas of the world, but each tide model has its own strengths and weaknesses.

A.3. Solid Earth pole tides and LOD tides

The effective solid Earth pole tide potential follows from differentiating the centrifugal potential Z (section 2) at a specific point at the Earth's surface (with the spherical geocentric coordinates $\theta,\ \lambda,\ r$) with respect to θ and the relevant Love numbers. The effective solid Earth pole tide potential, restricted to degree 2, is

$$Z_{\mathrm{P}} = -(1 + k_2 - h_2)\omega^2 r^2 \sin\theta\cos\theta\Delta\theta \qquad \text{(A.5)}$$

with the mean angular velocity ω of the Earth's rotation [29]. The position of the instantaneous rotation pole given by the polar motion $(x_{\mathrm{P}}, y_{\mathrm{P}})$ with respect to the conventional mean rotation pole $(\bar{x}_{\mathrm{P}}, \bar{y}_{\mathrm{P}})$ is described in [29] by

$$\Delta\theta(t) = (x_{\mathrm{P}} - \bar{x}_{\mathrm{P}})\cos\lambda - (y_{\mathrm{P}} - \bar{y}_{\mathrm{P}})\sin\lambda \qquad \text{(A.6)}$$

with the pole coordinates given in units of arc seconds.

In the same way, the effective potential due to variations in the angular velocity ω of the Earth can be determined, referred to as effective LOD tide potential. Differentiating the centrifugal potential with respect to ω leads to

$$Z_{\mathrm{LOD}} = (1 + k_2 - h_2)\omega r^2 \sin^2\theta\Delta\omega \qquad \text{(A.7)}$$

with

$$\Delta\omega(t) = \omega\frac{\Delta(\mathrm{TAI} - \mathrm{UT1})}{\Delta\mathrm{UTC}}, \qquad \text{(A.8)}$$

using the differences TAI minus universal time 1 (UT1) and $\Delta\mathrm{UTC} = 86\,400$ s. The difference TAI–UTC is 36 s since 30 June 2015, when the presently last leap second was introduced. The pole coordinates and $\Delta\mathrm{UT1} = \mathrm{UT1–UTC}$ are provided as Earth orientation parameters (EOP) by the international Earth rotation and reference systems service (IERS), e.g. within the long-term series EOP 08 C04 as daily values (http://hpiers.obspm.fr/eop-pc/products/combined/C04.html). For the computation of the solid Earth pole tide and the LOD tide potential, the program PREDICT is extended accordingly. The amplitude factors $(1 + k_2 - h_2)$ are set to 0.69. The pole tide and the LOD tide potential reflect centrifugal effects and have to be added to the static gravity potential in (5).

A.4. Non-tidal mass redistributions (atmosphere, oceans, hydrology)

For the estimation of the gravitational potential due to non-tidal mass redistributions in the atmosphere, the oceans and continental water storage, the spherical harmonic coefficients from GFZ AOD1B ATM and OCN [49, 50] as well as NASA GLDAS [55] are used, respectively. The normalized coefficients $\overline{C}_{nm}, \overline{S}_{nm}$ include the attraction of the varying masses and the indirect effect according to Love numbers k'_n within the spherical harmonic analysis, hence providing the total potential according to $(1 + k'_n)$. The total potential is computed for a point with spherical geocentric coordinates $(\theta,\ \lambda,\ r)$ by the spherical harmonic synthesis [42]

$$V^{\mathrm{total}} = \frac{\mathrm{GM}}{r}\sum_{n=2}^{n_{\max}}\left(\frac{a}{r}\right)^n\sum_{m=0}^{n}\overline{P}_{nm}(\cos\theta)(\overline{C}_{nm}\cos m\lambda + \overline{S}_{nm}\sin m\lambda), \qquad \text{(A.9)}$$

with the given global geopotential model parameters GM (geocentric gravitational constant) and a (equatorial radius). The radial displacements are computed from the same model by

$$\mathrm{d}r = \frac{\mathrm{GM}}{gr}\sum_{n=2}^{n_{\max}}\left(\frac{a}{r}\right)^n\frac{h'_n}{1 + k'_n}\sum_{m=0}^{n}(\overline{C}_{nm}\cos m\lambda + \overline{S}_{nm}\sin m\lambda) \qquad \text{(A.10)}$$

according to [82], while the effective potential follows from

$$V^{\mathrm{effective}} = \frac{\mathrm{GM}}{r}\sum_{n=2}^{n_{\max}}\left(\frac{a}{r}\right)^n\frac{1 + k'_n - h'_n}{1 + k'_n}\sum_{m=0}^{n}(\overline{C}_{nm}\cos m\lambda + \overline{S}_{nm}\sin m\lambda). \qquad \text{(A.11)}$$

The coefficients up to a maximum degree and order 100 and a time sampling of 6 h are available from GFZ as AOD1B product [49, 50]), which is routinely employed for the dealiasing of measurements from satellite gravity missions. The AOD1B product represents the contributions of the atmosphere (ATM) and the oceans (OCN). The data sets are available from the information systems and data center (ISDC) of GFZ (http://isdc.gfz-potsdam.de/) and routinely provided with a time delay of a few days.

For the analyses of large scale variations of the gravitational potential due to continental water storage changes, various products can be applied. Monthly GRACE solutions are available as spherical harmonic coefficients up to degree and order 90 from the International Centre for Global Earth

Models (ICGEM [83],). Present global hydrological models are, e.g. the Water Gap Hydrological Model (WGHM [84]), GLDAS (Global Land Data Assimilation [55]) or LSDM (Land Surface Discharge Model [85]). Due to the low frequency variations of the hydrological effects, a time resolution of a few hours is not required in this context and monthly mean values can be used instead. For the estimation of seasonal variations in the continental water storage, GLDAS monthly spherical harmonic coefficients for degrees and orders between 2 and 100 are used, which are available from the GGFC (IERS Global Geophysical Fluid Center; http://geophy.uni.lu/ggfc-hydrology.html). A wide range of various loading products is also provided by the EOST Loading Service (http://loading.u-strasbg.fr/), which is an official GGFC product.

The spherical harmonic expansions from the context of the GRACE satellite mission are well suited to assess the magnitude of the non-tidal effects in the atmosphere, oceans and continental water storage over the whole Earth with a spatial resolutions of 200 km ($n_{\max} = 100$). For an enhanced modelling at specific sites with higher resolution, the Green's function approach (convolution integral) can be utilized up to very high degrees [38]. However, the differences between the spherical harmonic and Green's function approaches do not affect the level of 10^{-18}.

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